

REKAXIME INJECTION

DESCRIPTION:

REKAXIME INJECTION 500 mg and 1000 mg are packed in 10 ml amber glass vials. The colour of the powder is white to cream coloured. Solution of the reconstituted product ranges from light yellow to amber.

COMPOSITION:

REKAXIME INJECTION 500 MG: Each vial contains Cefotaxime (as Cefotaxime Sodium) 500 mg

REKAXIME INJECTION 1000 MG: Each vial contains Cefotaxime (as Cefotaxime Sodium) 1000 mg

PHARMACODYNAMICS:

Cefotaxime has a bactericidal action similar to cephamandole, but a broader spectrum of activity. It is highly stable to hydrolysis by most beta-lactamases and has greater activity against Gram-negative bacteria than first- or second-generation cephalosporins. However, it generally has slightly less activity than first-generation cephalosporins against Gram-positive organisms.

PHARMACOKINETICS:

Cefotaxime is administered by injection as the sodium salt. It is rapidly absorbed after intramuscular injection and mean peak serum concentrations of 11.9 and 25.3 µg per mL have been reported about 30 minutes after the equivalent of 0.5 and 1 g of Cefotaxime, respectively. Immediately after the intravenous injection of 0.5, 1, or 2 g of Cefotaxime mean peak serum concentrations of 38, 102, 215 µg per mL, respectively, have been achieved with concentrations ranging from 1 to 3.27 µg per mL after 4 hours. The elimination half-life of cefotaxime is about one hour, but is increased in neonates and in patients with severe renal failure. About 40% of Cefotaxime in the circulation is reported to be bound to plasma proteins.

Cefotaxime is widely distributed in body tissues and fluids; therapeutic concentrations have been achieved in the cerebrospinal fluid when the meninges are inflamed. It diffuses across the placenta and is excreted in breast milk.

Cefotaxime is partially metabolised to desacetyl-cefotaxime, which has some activity, and to inactive metabolites. Cefotaxime is eliminated mainly by the kidneys and about 40 to 60% of a dose has been recovered unchanged in the urine within 24 hours; a further 20% is excreted as the desacetyl metabolite.

INDICATION:

Infections of the respiratory tract, including throat and nose; of the ear; of the kidneys and urinary tract; of the skin and soft tissues; of the bones and joints; of the genital organs, including gonorrhoea; of the abdominal region.

Septicaemia, bacterial endocarditis, meningitis; for perioperative prophylaxis in patients who are at increased risk from infection, and for the prophylaxis of infections in patients with reduced resistance.

RECOMMENDED DOSAGE:

Dosage, mode of administration and intervals between injections depend on the severity of the infection, sensitivity of the pathogens and condition of the patient.

Adult: The general dose for adult and children above 12 years is 1 gram every 12 hours up to 2 grams twice daily IM or IV. Maximum dose is 12 gram per day, IV or IM.

Infant up to 12 years: 50-100 mg/kg body weight daily, divided into several equal doses injected at intervals of 12-6 hrs. In some cases, life-threatening infections were treated with daily quantities of 150-200 mg/kg body weight, which were well tolerated with a maximum of 12 gram per day.

In premature infants, renal clearance is not yet fully developed; therefore, daily doses of 50 mg/kg body weight should not be exceeded.

For the perioperative prophylaxis infection, one of the above single doses is administered before the start of surgery. Depending on the risk of infection, the same dose may be repeated after an interval of 12 hrs.

For the treatment of gonorrhoea in adults, a single dose of one 1 gram IV or 500 mg vial cefotaxime should be administered IM. With less sensitive bacteria it may be necessary to increase the dose. Before the start of therapy, the patient should be examined for syphilis.

Children > 12 years old are treated with the adult doses, for which cefotaxime 1.0 g is available.

Dosage in patients with impaired renal function: In patients with a creatinine clearance < 5 mL/min the maintenance dose should be reduced to half the normal dose. The initial dose depends on the sensitivity of the pathogens and the severity of the infection.

These dosage recommendations are based on experience in adults.

Administration: Cefotaxime solutions should be administered as soon as they have been prepared. A pale yellowish colour of the solution does not mean an impairment of the antibiotic efficacy.

The duration of treatment depends on the patient's response. It must be continued for at least 3 days after normalization of the body temperature.

Reconstitution and diluent:

Rekaxime 500 mg:

For IV: Dissolve in at least 2 ml of water for injections and inject over a period of 3-5 mins. either directly into a vein or into the distal part of the clamped-off infusion tube.

For IM: In cases of gonorrhoea in adults. Dissolve a 500 mg vial in 2 ml of water for injections and then injected deep into the gluteus muscle.

Rekaxime 1000 mg:

For IV: Dissolve a vial of 1000 mg cefotaxime in at least 4 ml of water for injections and then injected over a period of 3-5 mins., either directly into a vein or into the distal part of a clamped-off infusion tube.

For IV infusion:

Short infusion - 2 vials of 1000 mg cefotaxime is dissolved in 40 ml of water for injections or a 10% glucose solution then administered over approximately 20 minutes.

For continuous drip - 2 vials of 1000 mg cefotaxime is dissolved in 100 ml of isotonic saline or glucose solution and infused in 50-60 mins.

For IM Injection: Dissolve the contents of a 1000 mg vial of cefotaxime in 4 ml of water for injections and then injected deep into the gluteus muscle. It is advisable not to inject > 4 ml into either side.

If daily dose exceeds 2 g of cefotaxime, IV injection is to be preferred.

Note: Sodium bicarbonate solutions must not be mixed with cefotaxime.

ROUTE OF ADMINISTRATION: For IM or IV injection

CONTRAINDICATIONS:

Cefotaxime should not be used in patients with known hypersensitivity to cephalosporins or with a history of a previous major allergic response to a penicillin, due to the possibility of cross-sensitivity.

WARNINGS & PRECAUTIONS:

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reaction) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with REKAXIME INJECTION, careful inquiry should be made concerning previous hypersensitivity reaction to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, REKAXIME INJECTION must be discontinued immediately and appropriate alternative therapy instituted.

INTERACTION WITH OTHER MEDICAMENTS:

Azlocillin. When cefotaxime was given in association with azlocillin, the mean terminal half-life of cefotaxime increased from 1.12 to 1.36 hours in patients with normal renal function and from 2.82 to 4.78 hours in patients with advance renal failure.

Mezlocillin. The clearance of Cefotaxime was greatly reduced when given with mezlocillin; mean clearance values for cefotaxime were 256.7 and 149.9 mL per minute when administered alone and with mezlocillin, respectively.

The activity of cefotaxime may be enhanced by aminoglycosides such as gentamicin.

Probenecid competes for renal tubular secretion with cefotaxime resulting in higher and prolonged plasma concentrations of cefotaxime and its desacetyl metabolite.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Cephalosporins cross the placenta. Adequate and well-controlled studies in humans have not been done. However, studies in animals have not shown that Cefotaxime cause adverse effects on the foetus.

Use in Lactation: Cephalosporins including Cefotaxime are distributed into breast milk, usually in low concentrations. However, problems in humans have not been documented to date.

SIDE EFFECTS:

The most frequent side-effects are rashes, phlebitis, diarrhoea and candidiasis. Vomiting; abdominal discomfort; headache; allergic reactions including rashes, pruritus, urticaria; serum sickness like reactions with rashes, fever arthralgia; erythema multiforme; toxic epidermal neurolysis; pseudomembranous colitis; hypoprothrombinemia; oral candidiasis; vaginitis; biliary sludge; renal disturbances and pseudolithiasis.

Adverse reactions: Rare but clinically important adverse effects are leucopenia, eosinophilia. One case each of thrombocytopenia and collapse after injection were reported.

Skin and subcutaneous tissue disorders: Frequency 'very rare': Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS).

SYMPTOMS AND TREATMENT OF OVERDOSE:

Emergency Measures to be Taken in the Event of Anaphylactic Shock: The following emergency measures are generally recommended:

At the first signs (sweating, nausea, cyanosis) interrupt the injections immediately but leave the venous cannula in place or perform venous cannulation. In addition to the usual emergency measures, ensure that the patient is kept flat with the legs raised and airways patent.

Emergency Drug Therapy: Immediate epinephrine (adrenaline) IV: Dilute 1 mL of commercially available epinephrine solution 1:1000 to 10mL.

In the first instance slowly inject 1 mL of this dilution (equivalent to 0.1 mg epinephrine) while monitoring pulse and blood pressure (watch for disturbances of cardiac rhythm). The administration of epinephrine may be repeated. The glucocorticoids IV, e.g. 250-1000 mg methylprednisolone.

The glucocorticoids administration may be repeated. Subsequently volume substitution IV, e.g. plasma expanders, human albumin, balanced electrolyte solution.

Other Therapeutic Measures: Artificial respiration, oxygen inhalation, calcium, antihistaminics.

INCOMPATIBILITIES:

Drugs: Admixture of β -lactam antibacterials, aminoglycosides and metronidazole may result in substantial inactivation. Should not be administered concurrently / mixed / injected at the same site as cefotaxime. Also, not to be mixed with these drugs, may form immediate precipitation / claudication / change in colour:

- allopurinol
- aminophylline
- doxapram
- fluconazole
- lidocaine
- filgrastim

Solutions: Lignocaine hydrochloride should not be used as a diluent for intramuscular injections in patients who are sensitive to lignocaine.

Cefotaxime is incompatible with alkaline solutions such as sodium bicarbonate and also incompatible with hetastarch sodium chloride.

STORAGE CONDITIONS:

Store below 30°C. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

REKAXIME INJECTION 500 MG: In box of 1 vial x 500mg, 25 vials x 500mg

REKAXIME INJECTION 1000 MG: In box of 1 vial x 1000mg, 25 vials x 1000 mg

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59 Kawasan 3,

Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.